

ActInfLab 2021 Quarterly Roundtable #4

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hmm [Music] hello and welcome everyone to the academ flab to our quarterly round table number four that was our first ever playing of a musical intro by d schwakoff so thank you for that and happy birthday to yvonne so welcome to this fourth lab meeting to our end of 2021 discussion not quite but almost welcome to activelab everyone we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this slide this is recorded and an archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams so just make sure to raise your hand in jitsi because no one will be called on but if you want to add anything please feel free just raise your hand in jitsi and we will get to it today the way that it's going to work we're going to address the lab scale we'll have introductions then we'll talk about some updates at the lab scale and then we'll go into spending most of the meeting discussing the updates at the organizational unit scale and then talk about the future but first we can say hello and just introduce ourselves i'm daniel i'm a researcher in california and i'll pass it first to dean hi my name is dean i'm in calgary canada and i'll pass it to david yep i'm in the mountains of the philippines and um pumped up that i can stop writing code in a new language and write code in another new language for a while preferably sumo yvonne hello my name is iran i'm in moscow it's uh i'm a researcher in system management school and an active conference club um alex or stephen hello i'm stephen i'm here in toronto um just in north york just north of toronto at the moment and i work a lot with um embodied sense making and spatial approaches and community development and i'll pass it over to alex thanks yes i'm alex i'm in moscow russia and a researcher in active inference and system thinking indulges so thanks thank you and rj if you would like to say hi yeah sure uh i'm rj i'm in new york in the u.s uh i'm a non-resident fellow at the atlantic council and i do research on sense making and knowledge management well thanks everyone for joining this will be a fun discussion maybe some more will join in um and anybody who's watching live can just ask a question in the live chat we're going to start with the updates at the lab scale so actin flab updates we started this

year with a interactive and participatory mission slash vision slash project which was to ask whether active inference could develop into a community and a field and all that comes along with that while also drawing on the best practices that we saw elsewhere especially in systems engineering community of practice that yvonne and alex had a lot of familiarity with and other experiences that we and every participant brings to the table uh having a community for active inference that was participatory and accessible for those with many different backgrounds and trainings having a body of knowledge that includes courses and competencies things we'll talk about soon and also reflected the nature of modern work and professionalism and collaboration so these are just a few of the things that we wrote down long long ago and it's still a living document so please come get involved all actions are participatory stephen i was thinking we could also say a community of communities a nested set of uh communities uh inside a community that overlap in the kind of markovian sense yes let's graphically look at that so this is a little updated graphic and i hope it looks good for everyone so on the right side in the gray background we have the structure of the active lab itself so this is the lab scale that we're addressing in this section we have three primary organizational units which are coms communications edu education and tools tools we also have dot admin but they won't be presenting on projects today also in the active lab we have some of the pieces of the puzzle that we have gleaned from systems engineering community and other areas and we're going to be talking more about the advisory board soon and about other initiatives that we're undertaking so that's sort of inside the lab this is the lab scale under the lab blanket we have our organizational units dot coms is outputting different products into the broader ecosystem like externalities or runoff and so that's kind of this teal scroll image and coms is really our primary interface with the external ecosystem we can think about this external ecosystem as containing all kinds of different things in it so there's artifacts there's texts like papers and videos and all kinds of other uh documents and artifacts and then there's also people and their skills and their connections so we can start in with this red bounded box of the current body of relevant publications which as we've seen stretches back and forward and left and right so there's a lot of relevant publications but there is a relevant body of publications and we have used the live stream format kind of like a journal club to go into the publications themselves each live stream is centered around a publication which we'll talk more about in comms comms produces a guest stream which bridges from that circle of established knowledge and people into broader interactions amongst participants in the topics that they're interested in like a two-way street between let's just say active inference and field x so some people are going to

be hearing about active inference for the first time others are going to be inside of active inference hearing about x for the first time and then we look forward to having increasingly broad avenues to communicate like org stream that help to bridge even one step further and outside of the realm of just research and development may be thinking more about practice and all the different ways that people around the world are engaged in practices every day that may be related to or influenced by active inference so here's kind of this nested of uh circles like an onion around the active inference literature sensu strictu and then all the ways that our outputs into the ecosystem help to scaffold and modify and catalyze sense making in that environment which can be overwhelming also which we uh picked up on from the systems engineering practices and community organization we've engaged in several development initiatives which we'll talk more about in the sections that are relevant but the ontology working group is uh upskilling and learning about ontology and the systems thinking reading discussion group is doing similar for systems thinking so we'll talk more about those but this is just our initial forays into some of the kinds of products and lab functions that are important and we hope serve a positive role inside the lab and effect a positive outcome outside the lab in our local niche so there's our internal states over on the right side and then there's our niche modification products in the left side any comments that somebody would like to add here uh stephen yeah i suppose i like to think about how the ontology working group is is hopefully helping to bridge and fill a gap in the in the complexity of all the different ways that active inference gets used so it's kind of hopefully going to help for that applied active inference to be more multi-disciplinary interdisciplinary transdisciplinary no small task but i think it's a useful endeavor thank you agreed and there's the ontology itself um as a product in the niche so multiple representations are very important for distributed systems for complex systems this is kind of a snapshot visual layout but we can also look at more of a narrative overview so just to recap during 2020 viatgan at all that's alex and others including many on this discussion now collaborated on a paper that was released as a pre-print on september 9th 2020 called active inference and behavior engineering for teams so this is as close to a prerequisite for understanding the narrative and the context of the actin flab as exists at this point after completing that paper we sent out an initial call for collaboration with an active inference lab and that was at the end of 2020. during 2021 we engaged starting at the beginning of january in our first year of active lab activities many of which we're going to be talking about today that's kind of our recap moment um and then we're going to also be looking forward to 2022. so this is like a like a sandwich we're going to have the two pieces of bread right now and then

that will be our appetizer and then we're gonna spend most of it on the meat itself so it's an odd sandwich but please what do you expect in 2022 we're going to do a few things similarly and a few things differently so on the similar but different wavelength we are going to be continuing to implement our projects and plans within each organizational unit so that's what we're going to discuss today in the organizational unit sections it's what we talk about every week when we meet in the organizational units so that kind homeostatic or functional operation of the lab is the substance of what we're actually doing and so that is going to continue we're also bringing a few new ideas and structures to the table and practices so one is the inaugural advisory board cohort which we'll talk about on the next slide and then a little bit on the um not having their own slide of their own but important ideas that we look forward and call for participation around formalizing the lab governance there have been so many important and interesting governance questions that have arisen during the past year as well as everyone experiencing similar challenges in their own located in online teams so that's something we want to understand better and then also as we learn more and embody and enact decentralized science or desai everything that comes along with this from who's doing research what questions are they asking how is it funded how is it communicated how is it verified the whole research stack is evolving so welcome to the 2020s and we want to understand how we as a lab can take the best aspects that are emerging from this domain of desai while not getting lost in the technicalities or committing grievous ecological errors anything else that anyone wants to address these were just a few of the kind of top level points that we all worked on when we were preparing the slides dean yeah i think this is a fantastic summary and i think for people who are kind of new to this or maybe are still constantly sort of evolving their basic understanding of this um i think what we're what we're seeing right here on this slide the tension between a lot of constant updating which is which is kind of what the active inference tool um enables and then the piece about okay so these are the things that we've established the kind of top-down consistent constant stable things that we think um have some shelf life and can carry forward so i think by showing this sort of chronologically we're sharing with the audience the fact that there is a tension there isn't one constant that is going to take over and there isn't a constant updating mean we're always throwing out the baby with the bathwater and i think you may not necessarily hear or see that in a timeline but i think that's front and center in terms of how we organize ourselves we organize ourselves with the with the intention of remaining in a in a in a tension that constant updating versus what's stable and is carried forward thanks dean and welcome blue great point and it is like the nesting of

models in active inference where there's a slower tick at the deeper scale of the model so every year there's 12 ticks for months and then within a month there's more and more and that takes us to the advisory board which is something structurally that we're introducing in 2022 this lab advisory board first we can say it's not a formal voting or governance hashtag not legal advice not financial advice they're not our managers it's a advisory board the advisory board is helping to move the temporal scale of the labs organizing from the kind of day-to-day sometimes it felt like we were just putting out little surprises all the time to the annual scale and beyond so as we think about deepening the deepest level of our model for vision and for alignment while also refining the finer points in our model that's kind of like the icicating is like the micro scale and then we have deeper levels like semantics of vision so that's kind of the way that we think about these temporal scales and we're already in the process of onboarding individuals with expertise in a variety of important domains but please contact us if you have any interest before say the middle of december of 21 and you might want to be involved in this inaugural cohort of the annual role at this point for the advisory board or we'll be there to onboard you in 23. what we do when we onboard them in the advisory board just to be clear and because many of them will be listening to this discussion first we make personal contacts with organizers and have consensus amongst lab organizers about that invitation and we'll have more governance for that future we then have a live meeting with a potential board member and assign them to one or more organizational units so no side doors no free floating i'm here but where everybody is in an organizational unit and then we describe the specific synchronous and asynchronous affordances and roles and requirements that the advisory board members will engage in so it's not just an open-ended ask for you to do work it's actually a very structured conversation that will hopefully be win-win so we're really looking forward to the advisory board's service does anyone want to add any comments on the advisory board and we'll probably hear more from them once we've confirmed the personnel but right now it's just a um we're still in the onboarding phase okay any other lab scale comments before we go into the organizational units okay our first organizational unit as a educational nonprofit is edu education what we're going to do for each of the organizational units is we've just copied from our coda document which we'll talk more about we've just copied the ongoing projects so the idea is these kinds of single source of truth and centralized ways of working together allow us to have an asynchronous lab and then also when we need to present and share we just take screenshots or copy out tables so that keeps it really efficient these are the six projects or five projects that we have in edu so these are not the total scope of all the projects happening by

participants these are within the organizational units markup blanket the goal of edu is to scaffold the participatory and dynamic active inference body of knowledge so these five projects are the current active threads and if you see a space to improve one of these or add another one then get involved so on to the first one of these projects this is the fristen symposium transcript so i'll summarize it and then dave if you'd like to add a comment or anyone else so in june 2021 we had our first applied active inference symposium with professor carl fristen and each of the three organizational units had about a one hour session the videos are watchable and many insightful things were said and discussed so the idea of this project is to use automated speech-to-text processing plus manual copy editing and enrichment of those texts to get an accurate transcript of the discussions so that we can upload it as a preprint get a doi and make it citable so instead of oh i think somewhere somebody said this thing let's have in-text something we can copy out and connect core terms to translations use it for the captioning of the youtube videos for accessibility and then looking beyond this one important symposium we had this year towards transcripts for maybe all live streams and integration more thoroughly with the ontology so translation of terms linking to other knowledge resources so that's the fristen symposium transcript dave or anyone else do you want to add a comment yeah um the three transcripts are um in daniels and blues and some other people's hands and um as much markup as folks want to do that's fine if you want to just take what i've got and make them perfect and get all the cross references and so on that's great if you want to just um mark up uh the change of speakers and maybe throw in some obvious paragraph terminations just empty lines that's that's fine um and i have the option later on of just doing some some automated cleanup uh trying to find end of sentences that aren't marked otherwise but um you know i think this is going to be somewhat quicker than just taking the raw youtube output well we shall see uh next year after some other things i'm going to go in and improve the program a lot more do the stuff that i thought i would have finished by now but you know when i can't change the parser and i have to live with the parts that i've gotten it's only got half of the vocabulary and i gotta get around that it takes longer so that's good if you can't pass the one you want parse the one you have steven yeah i was thinking that one thing to think about when we have these talks with um kristen and these presentations is we're able to as well as clarify the stuff which is um already possible to do and separate it from what is um you know is beyond our knowledge at the moment we get this zone of proximal development the vaikovsky which our russian friends will be very fond of um so that idea of where we can be we get a sense of where the sort of the extension of what the established

knowledge might be and i think that's also a useful thing to clarify um because that's also part of where this applications are going to be good point we're not just recapping memorized facts that's the first in symposium transcript the second project in edu is the ontology working group so in ontology working group we are reading this textbook by adam peace the ontology uh a practical guide textbook and the bigger picture here is that there's a continuum of formality and expressivity of what is called ontology ranging on the left side from mere terms lists through more structured and enriched terms lists like structured glossaries and informed hierarchies verging towards more formal data structures that allow increasing computational power in many different dimensions and over here the sun rising in the east of sumo and uh sumo's the language and the framework that's discussed in the textbook suggested upper merged ontology so that's what we're learning about in the ontology working group and we meet every two weeks we discuss what we've read and worked through the exercises together over the course of the year the ontology working group has been kind of pushing like a little train towards further and further right from just the terms list that we started at in the beginning of the year to a lot more which we'll talk about soon but the ontology working group is specifically those who are learning ontology by doing and reading the textbook any thoughts or comments on ontology working group and we'll talk about the ontology more broadly in just a few slides so another development initiative which um i'll allow yvonne to summarize is the systems thinking reading and discussion group so yvonne maybe you could summarize the intention or the background here yes thank you well in initially the system thinking course was represented by system management school in moscow and we decided to provide a common language for the lab all of us from different domains with different perspectives and we decided that it will be somewhere we can one another better here i gave some small example when where one object can describe a totally different one person can hear can see here a face and another vest and once all of them talk about leaps they totally to miss understanding and to prevent this um situation this group could you could you texts next yes yes there is the interface of the course is this is uh online course based on iso standards and for documents that include system approach in engineering and in different a lot of fields of course the course has a list of concepts that you can actually interconnect it with the active inference concepts every every participant can now uh upload it to the course in appropriate way start to learning system and thinking as system management school providers of course has eight sections now a lot of tasks check your understanding you can find all information about the course on systemschool.com thank you yvonne and also you and others translated it

from the russian to this english course which were beta testing so it's being offered for free and we provide them feedback so it's a cool relationship between our groups that helps benefit our participants any comments on st rdg one of the main projects in edu something we spend half or more of each meeting on is the actual active inference ontology itself so here we're showing um just the top page of the working active inference ontology we released versions um more stable versions earlier in the year we're working through the core terms and then for each core term getting a bunch of references and quotable citable definitions so not saying that they're the truth just how it was used you know a system is a or blanket states are each week we have a discussion on these terms one term per discussion and we kind of go deep and hear many different perspectives and gather more resources and then we work towards proposed synthetic definitions that synthesize in the term of synthetic not artificial that bring together a few of the senses or clarify is there just one sense or are there multiple senses that the term is being used and then the terms themselves as well as how the terms are connected to one another will form some of the early versions of our educational materials so how about that five minute video that everyone has always wanted about what is a markov blanket or that five minute video about how is a generative model related to epistemic and pragmatic value those are the kinds of questions that we want to have like a video but more importantly this deep knowledge linking underlying so just that's the first cover sheet and then i'll get to you steven um the second sheet has translations of those core terms into human languages so we have russian portuguese spanish french and italian very well completed we're definitely looking for more languages to be added so if you're speaking or learning any language then it would be awesome even just to do two or five words terms because that helps modify the niche so that stigmatically another language speaker may be able to pick up where you left off so this would be extremely helpful and important global work for somebody who speaks one of these languages or any language to start to pin together just with the terms at first this active inference corpus which is primarily written in english as a natural language mathematics and computer languages as technical languages and so we'll be able operate better across those different human and technical languages and then the last sheets of this tab here just show the supplemental terms and equations so the equations that's going to be a long game with corralling the formalism and how that has evolved and how different letters have meant different things through different times but then also supplemental terms there's so many fun terms that come up in our discussions but they're not necessarily core or they may be we may update them and put them in the core later but things that people bring up that like are cool

and important to learn about and link around we can add that to the the term supplement so we kind of get to have a big gutter with a term supplement while also having a very curated core terms list that will be the basis of education so stephen yeah i'd just like to also thank again that ivan and alex there for bringing in that system's work i think it's really valuable and i think this as you mentioned there the different languages are also useful because even different languages have different ways of thinking about time they have different cultural so if we're looking at pluralism um but also trying to clarify and get ways to have common languages this is going to be actually extends that and one other thing i'd like to mention is that it can build the bridge with the field of coaching i think that coaching systemic coaching and ontological coaching are two very fast growing fields and i know from the literature that they're really moving into looking at um pluralism and different types of social construction and i think so this this before it's a bit of a bridge too far but i think with this work as well from alex and ivan and this other work with the ontology is starting to make it not bridge too far hopefully it's it's it's spannable very cool the last project to discuss in edu is course development itself you know what people expect of a edu to engage in and we know this is one of the most important projects and we also knew that ontology not necessarily the practice of ontology itself but this core term ontology would be the backbone of the course broadly in this project were interested in how to provide the affordance for people to learn and apply and communicate active inference we had great discussions during the year on live streams and in edu about instructionism and interactionism and how do we prevent this from just being a sort of plug and chug course how do we enable interactions and his own approximate development like was being described and this tension that dean described earlier as well between the sort of the technical the minute particulars and the big picture and the unknown and the thirdness and the betweenness and um with a special focus on applied active inference and also enabling this education for individuals of many different backgrounds inside academia outside academia there's a whole world out there and there are not paths from here to there so that's what course development is going to be exploring in 2022 and beyond especially everyone is welcome to get involved with all projects but in edu if knowledge engineering sounds interesting to you whether it's the translations or diving into the literature to find definitions and references bringing your experience on curriculum or learning by doing on the ontology those capture a few of the projects that we talked about here and that's some of the main practices that are involved in edu any dot edu thoughts oh wait unmuted then continue sorry about that there's some cultures we're taking a very western-centric approach to this which is

fine we have to start somewhere and i think that there are some cultures that simply won't have words or verbs that describe some of the things that happen in their particular niche that those with a western view don't necessarily have the terminology for and i think we're keeping that aspect of this open when we're trying to figure out whether or not there can be a bridge from calgary to moscow or if there are other ways of being able to transport ourselves and migrate that are better than trying to put a solid structure in so sometimes when we're working on this ontology piece we might only be able to collapse to two or three different ways of describing two faces and a vase and i think that's one of the other things we we want to let people understand is that there is no one singular answer to this that when active inference is actually being used we distribute and there are going to be times when we can agree to disagree on how to describe something thank you dean it's like there's what was said in first in 2013 the paper there's how you just said it right now there's how blue's gonna say it in six months and those are like observations those are not the latent hidden state even if they all concur that isn't alone evidence that that is the hidden state so we actually can take that kind of signal processing perspective and integrate it with ideas like pluralism and distributed systems to talk about how to have inclusive so in comms the second organizational unit we are interested in communication the goals of this organizational unit are to organize the internal lab project and activities kind of the connective tissue circulatory system and also to carry out all forms of communication with external entities so if.edu needs to send an email they can just pass it to coms and say please email these nine people with this information thank you and dot coms loves emails the first project well there's uh six projects on here live streams social media podcast end of 2021 treat monthly newsletter and updating the coda and the playbook so first on the live streams of which we're in one now we had a fun year of live streams the live streams technically began before octave lab in july 2020 but we have um multiple series of live streams and we've done well over 100 total 85 around papers and then 14 guest streams presentations seven model streams and three math streams which are more technical and then um roundtable meetings like this one orgstream is a new series that just began we have one video in there about organizational design practice and then the applied active inference symposium lest anyone think the year is over we still have some upcoming papers to read for 32 still cast the chaos markov blankets in the first two weeks of november 33 thinking like a state with avel serval 33 in the last two weeks of november and then we'll have two more papers probably that we'll discuss in december but just haven't set them yet so we'll also be probably moving this to coda having a nice way to look at the past and present live but for

now if you want to give a live stream like give a presentation and or discussion for a guest stream on your research or your thinking or your application just let us know and also if you want to suggest somebody who would be good for a live stream whether it's just like i read this paper could you reach out to this person or whether it's somebody you know and you'd like to co-organize the event or just make the connect we're planning for 2022 so that would be great if anyone listening has any thoughts or wants to present social media broadly just to give two updates we have over a thousand twitter followers and uh over 600 subscribers on youtube the slow and steady accumulation there on the podcast front uh then i'll pass this to blue so anything you want to add on social media blue and then please describe the podcast sure so just also on the social media front um i just kind of took over uh the facebook page and um or or took responsibility for you know promoting the actin flab on facebook so if you're on facebook um find us there because that's brand new and we have like i don't know a whole 10 likes on our page or something like that but but we've hooked up the podcast which will feed directly into the facebook page so if you get updates that way that's helpful and we'll try to post the live streams on there as well yeah new social media venue is always good we're also on discord right so um if you're on discord and and do that too um there's more social media ways to get a hold of us and then podcast i think we're up to like 30 episodes of the podcast at this point and we're starting to codify them um with links to each episode and the associated live streams and maybe in some future rendition of the website we'll be able to make all that information publicly available so that people can see who participated and what the topics are because frequently the podcast will just be like the markov blanket um even though the paper is a much broader paper or topic so the podcasts are coming along and yeah get in touch i mean there's always room for people to participate i've been doing a lot of them um also we're looking for music so if you want to submit some musical riffs to us uh for evaluation um definitely that's something on my on my to-do list is to like get an intro lead for for the lab podcast and if if you're interested in podcasting or audio production at all um get in touch it's been i've just kind of learned and figured it out but i'm happy to you know work with people show people the the ins and outs of that so yeah awesome job with these edits blue because sometimes the discussion can have even when it has just a few core themes which sometimes happens it can be interleaved complexly so it takes skill to untangle but the podcast is fun um number four the end of 2021 treat so this is just a little gift that blue and i wanted to send out so this is kind of our project we just thought um we could send people some physical stickers and potentially send a non-fungible token an nft purely as a gift no value or utility so both of

these reflect the expansion of the lab the physical space as well as the digital and the cryptographic space we do think that cryptographic affordances will be very important for the future of science art and communities in web 3 and beyond we also want it to be mindful ecologically and meaningful for participants so if you'd like to be involved in this area of kind of taos and web3 and crypto and desai please come get involved with active lab because there's a lot to learn and help us do here just very briefly we have a monthly newsletter so that's um you can find that on our site activeinference.org and find other places to subscribe but this is a kind of monthly and then a few other special affordances but mainly just a monthly summary of our operations so good low traffic totally not spam way to stay updated and then the last comms project is um the live stream playbook so we do have a slides-based playbook for uh live streaming so i'll just kind of like show it here this is just for participation guidelines this is um checklists for roles participant checklist presenter checklist broadcaster checklist facilitator checklist organizer checklist so that was something that we developed over this year and we're looking to move that over to coda so that it can be implemented much more smoothly with automations the playbook itself from the technical details to the behavioral and the social and then also another coda related improvement project is going from the list that spreadsheet of live streams to this enriched table where we're gonna be able to add in keywords links participants um you see some upcoming streams here so it will be possible to very easily ask a question like i want to know what dave had to say about predictive processing before 2022 that will be a simple query to make and when we include the speech to text and have full text transcripts not just enriched tagging that'll also be very powerful but this will be really a fun project and it's also little micro tasks if anybody wants to contribute asynchronously like i loved number 14.2 i'm listening to it already i'm going to write down a few keywords that were spoken that's a huge way to help the lab think about how many times a podcast or live stream is listened to and no one enriches it or makes it searchable so this will help improve that situation connect these resources to each other so if you want to get involved in dot coms if you like communication and you are familiar with active inference or not of course all activities are open to everyone special roles or practices that could help a lot would be somebody who wants to like lead and grow and automate active lab social media accounts because that's not really our jam also if you want to get involved in co-organizing or facilitating or broadcasting live streams introducing new series compiling and curating research products onboarding participants new kinds of communications like blogs short writings tutorials there's so many things that we can do in comms and if you just think that's a

communication i expected to have existed i i expect that five minute video on markov blankets to have existed or i would prefer it existed we'll connect you with the affordance so that you can reduce your free energy by doing it and also helping people any comments on comms okay final organizational unit dot tools so in tools the overall goal is to enable the effective tool and instrument use for all active lab processes the second goal is to explore and design affordances for our niche resulting in effective action as well as innovations and tool development so it's a lot like comms there's this interior connective tissue role and then there's an external side in tools the internal side is our tools what we use in the lab and that's what the participants are engaged in because we're all participants here and then there's like this medium and deeper question what do active inference tools look like recommendation engines and robotics and all these kinds of things and we have explored that in some depth during the year but let's go over the projects themselves so first is the cloud infrastructure so this is an internal but eventually we'll have an external component as well we're looking to develop accessible and powerful computational resources for the there are several use cases that are very important some of them are admin tasks like having the website emails file storage crm like personnel management in um a secure cloud environment there's also use cases like a personalized learning dashboard or workspace so somebody can just log into their active lab accounts and have their resources and learnings right there for them in a high attention and high trust setting two special use cases that are can be mentioned here one is ontology development and natural language processing so sumo it's an ontology but it's actually written like a text program kind of like lisp so having an environment with sigma and a few other pieces will allow people who might not be able to set up these often esoteric programs on their own computer will have the cloud image virtualized and ready so that you can play around in this sandbox where all the toys work and then similarly on the machine learning side although much of early work in this vein was done in matlab in the spm statistical parametric mapping toolbox increasingly modern development is happening using the languages of python and julia and we spoke with people in live streams and guest streams who talked a lot about that and so we want to have a what's a fairly common setup in industry and in computer science research labs to have a cloud interface with for example interactable notebooks that helps you do machine learning in the cloud without having hardware or software installed locally so that's kind of the cloud infrastructure side and of course these aren't done projects these are things that people are involved in and can be involved in feel free to just raise hands if you have any other thoughts but um another project is the actin agent development so we here are

curious with what does the future of active software agents look like can we make an interoperable software package for active inference agent modeling and where could that be applied we started with the nethack challenge which we didn't muster the regime of attention for this year but it was ambitious and we had many many good discussions and we made a lot of useful advances we just didn't end up doing this very challenging machine learning competition there'll be so many cool areas to apply active inference to so that's kind of like just like people expect the edu to make courses but it does a ton of other stuff too tools sounds like it should be making something like this and we'll do other stuff too project three in tools is the participant onboarding um yvonne do you wanna add any detail here yeah as then already mentioned we this quarter extremely more use code tools and here we as other our activities we model a person who can simply be involved in different projects within the lab our activities we based on systems here different scale of the lab of project and the tasks within the project we have is not not a simple but we have a model that we use to to simplify the onboarding so within the project we overall and we opened to assign this role to performer with the skill we understand we need to make this role each role has been in organizational unit of the lab and to perform the task we provide some some information to the information to to have understandable understandable task any any skill you you have already thank you yvonne that relates to the crm integration which is usually using to refer to customer relationship manager but they're kind of like personal relationships it's a project that is on coda it's under the dot coms but the integration is a tool's concern and so just like yvonne laid out here using these ontologies for projects much of which are derived from systems engineering and systems thinking we'll be able to have everybody entered into the crm and then it'll say okay blue and i are going to be doing this live stream they added us to a calendar the template is here the papers here the emails in the inbox that will allow um very flexible coordination and also improve the accessibility because not everybody is going to be trying to um prune every part of the garden each person will be able to have a targeted regime of attention that is according to their preferences and also will be contributing to something bigger so these are projects involving making the organizational unit engagement more accessible project five uh yvonne again please describe what is the uh systems thinking reading and discussion group which was an edu project on the learning capacity what does it have to do with application in the dot tools as each project where we can we have to divide the project itself and the project that enable this project so to provide all participants with all needed instruments we need to enable it and uh more understanding of the course and the discipline of system thinking uh we

decided to set up reading and discuss group one per two weeks we plan to meet and discuss what we learn and what hard things we met in the course uh and here i provided two two screenshots it's that the first first of it is how it looks like in coda and the second one is the backlog of the course if body during the learning report issue on what typos typos it the course has or uh any comments they would like to and if you move to the next slide please yes so once once a student record an issue we have this table and each issue automatically placed to this this table because of different translators work worked on the course chapters different translators worked on different chapters and each issue is assigned to a special translator and within the systems system school we we fix and what what do we need to fix in the course thank you and this will be one of the kind of templates and inspirations for the active course that will be using tools that are state of the art which were selected from a rigorous process for the systems thinking course a sixth project for those who watch all the way to the end of lab meetings you know who you are is the graphical interface and extensible knowledge graph project so this is going to be something pretty cool heading into 2022 and beyond on the left side on the left side is an image that i'll just expand so it's a small this is matlab which is a proprietary program that you have to pay to have a license many academics are given access but that's not everybody within matlab the program there's a package called spm statistical parametric mapping which has been developed by professor carl firsten and many others for the previous decades inside of spm you'll find a function called dem demo and it will bring up this graphical interface which has kind of sundry links to some simulations some code examples some visualizations some papers but much of it is within matlab itself so this is kind of a resource that's been developed and locally optimized within this intra spm intra matlab format however one of our major projects for the lab and the community moving forward is going to be reimagining and deploying a graphical interface and an extensible knowledge graph for active inference so what does it look like to make a graphical interface an extensible knowledge graph that is participatory with open source development and modern affordances for knowledge graphs that uses the active ontology as a basis of the knowledge graph so for example here it's sorted by keywords but you know what do you do when one button fits under two keywords or two keywords apply to one button that's why this is a local optimization because to have that kind of advanced and even personalized re-rendering it's required to have a more advanced ontological system than just terms or principled hierarchies which is exactly why ontology working group has been involved in this work and these practices because the extensible knowledge graph starts coming into play on the far right of that spectrum we also

wonder if it can be deployed in a website so not requiring you to have matlab install spm find dem and then maybe it works or maybe it doesn't how about a website where there's maybe even computational resources so people can run simulations and save their own versions of the code in their own workspace and of course what else this is a project that we're barely just beginning so if you're interested in user interface or education or tool development something that would be an ontologically scaffolded knowledge graph containing very heterogeneous resources like terms and educational courses papers videos etc this could be a very important and also very active project so if that's kind of sounding cool to you then overall uh dot tools has all kinds of uh well we'll just delete this dot tools has ways for you to get involved if you'd like to contribute in any way let's go to the future and then of course everybody please feel free to raise your hand so the future section is short because we can actually just break it down systemically and systematically to use some vocabulary that i learned from systems thinking into these different levels of organization of our lab there are the next step steps for the as an entity which involves for example our nonprofit registration continue continuity and the onboarding of the inaugural cohort of the advisory board there's the next steps for the units inside of each lab which were discussed for most of the meeting today there's the next steps for the projects and the teams that are inside of each unit which have specific coda spaces and timelines and roles and requirements of themselves and then there's the next step for you as a participant if you don't know that's perfect because that means that you'll have a lot to learn by doing and reduce your uncertainty through action so if you're listening to this or if you're aware of it you're already a participant you're engaging in the regime of attention regimes of expectations related to acting of lab and active inference more broadly if you're here you're a participant if you know what your next steps are what policies you're going to select in the near future or now that's awesome if you do not know what your next action states should be what policies you should select or maybe even what policies you can select what your affordances are then you can get involved by just communicating with us so anyone can feel free to contact welcome scott to the very tail end um we're literally just closing up and uh just have about to have an open discussion where anyone can just add any points what was something that they liked about 2021 what's something they want to change what are they looking forward to in 2022 anyone's welcome here well i'm too i'm late to the game so i'm i don't know what was said before but i um i feel like the we're on the cusp um of folks benefiting from this kind of framing in a much more broadly and i think that was interesting i just saw on twitter there was an interesting article on the rethinking of pristine um pearl

blankets versus fristen blankets discussion which i thought was interesting and it's not i haven't read the article yet but it's interesting that notion of the maturation of the concept and whether or not that critique is accurate the critiques are helpful because they start to bound um framing and make it more useful for more people because it's not as over enthusiastic in its ambiguity on the at the edges and so i'm really excited about the maturation of both this concept and the communities that cluster around it and i feel like this particular initiative is going to be really pivotal in that and it's um i won't use the word priesthood and priestess hood here but um there's a lot of um evangelization that will be done in the in the next uh or evangelizing i guess that will be done in the next year and i'm really looking forward to it i think a lot of people are going to find that when they enter the space that people are are for some reason seem fairly reasonable given a lot of other spaces and willing to entertain the limitations and constraints what we have here because i think it is something that has a core of something that's new and offers helpful insights so i'm really excited about the coming year thanks yeah i think my takeaway from the months that i've participated so far is that one of the big differences kind of a dramatic difference if you will is that i find that working in this medium it's more like a cnc mill than a than a printer a 3d printer because instead of a bunch of building up we're essentially removing a lot of noise and trying to get to the idea of what are some of the signals that we otherwise wouldn't be able to see because we have a chance to interact the way that we have and do i don't think it's uh i don't know that it's necessarily a consensus building platform but i do know that it is certainly a way of being able to cross pollinate we don't tend to collapse to one particular type here and that that last input being a perfect example of what we're trying i think what we're achieving here and i think that the going forward piece really interesting because it isn't always describable in this moment i think a year from now when when we're doing our fourth quarter a year from now god willing um i think again the updates will tell us a whole bunch about what we learned in the meantime instead of um trying to come to a particular specific outcome and i think that's probably one of the greatest joys of participating in this thanks dean fellow chitzer with a raised hand all that sorry somebody who's muted but i'm not sure who exactly it is is it me you might yeah okay um yeah i i following on from what was said there is the idea of um the blanket early on was kind of like a part of active inference but the having been talking a lot more and you know thinking about the fristonian blanket as a particular way of thinking about blankets and knowing that these blankets are used in other contexts i think has helped mature things i mean scott's been talking about the idea of maybe synthetic blankets in other contexts and i think what's quite

interesting is i think the around the nature of blankets is going to be um as a tool or maybe as part of the ontology is going to help at least help separate out the discourse and i think that's not a bad thing because uh there's there's there's um there's there's uh there's markov chains and there's lots of these processes are used in other fields which are allied but are not active inference um and then well what happens to these when we come into that field and uh i think it also then starts to help with some of the pluralistic ontological questions allowing other parts to maybe be more formalized and then the ontologies might happen more through the way that we think about the realization of the blanket form and the sensory states and the observation states been interacting with thank you stephen so anyone else can uh raise their hand definitely the core terms yup scott go for it so you know it's interesting steven just got me thinking so one of the things we're doing this nsf thing some of you know about it right now and doing it with some other folks in involved in complexity and um we're doing virtual uh excuse me uh verified information environments and you got me thinking stephen you know credit card system we say is a verified information environment we're defining it as a thing that has business operating legal technical and social variables that give it integrity so it's interesting when you think about blankets in a way a verified information environment what we're saying is there's an externality or an ex there's a an external environment that can be named and characterized so it's like before there were countries nation-states we didn't say they're nation-states we had the feudal kingdoms or whatever and it was a characterization of a set of and it became paradigmatic only through broad adoption so we say now there's a society or we can say there's a museum or there's a school each one has a concept but it's really a set of relationships not a building and it feels like i wonder if markov blankets it just started feeling like perhaps there's a naming of a set of interactions a blob of interactions out there that can start to build up its own um not not rigor statistically but cultural um recognition and and if and that i wonder whether markov blankets are candidates for that i don't know if it's powerful enough a concept to me it feels like it may be but and and that set of practices that occur associated with markov blankets um i think can be elevated into best practices standards and institutions so we might actually see is the in markov blankets for instance might become institutionalized into fiduciary layers for representing parties things like that anyway you got me thinking stephen on that one nice thank you um i was just going to say that the core terms each of them are so rich and uh the opportunity to go into technical details and discussions as well as apply discussions there's so much more that could be said the only thing i'll say is personally i don't think that having names associated with terms is helpful because it's not

referring to a person themselves and so i think once we move beyond some of the name based nomenclature will open up into a more accessible and rigorous space not based around whether darwin really thought this way or not about evolution or whether markov which markov really thought this way or not about statistics we can just frame positively indefinitely and probably very austere what we actually mean and that will enable translation automation and many new things that just last names don't give but that's going to be part of the fun i never even thought of the mark of part as and everything occurred to me it was guy's name but of course it is but so i i vote for snugly we should call it a snugly as a generic term i think let's change all that into russian what's a russian what's russian for a schnoogley blanket what about snuggie like the blanket with arms it's already trademarked let's change that frequency to snugly um well um and then also one just mark off no just we are actually unsure which markov is being referenced this was some scholarship by dave to actually address the question which a markov the the elder or the younger are some of these terms referencing in the english-speaking literature and that is not always clear so there are just opportunities for international scholarship that we discover when we're curious stephen of course you've got the ian we can add on so the fristonian i mean it is problematic being his name but or the markovian um i mean that sort of opens it up to the hand waving but it um that may also sort of show that there's more than one version because you know you just hear markov blank it does to someone who came for the first time they may think well that's that's the thing it's not it's that's it right whereas um once you get into markovian monism and even that maybe that there's i can see why carl wouldn't want to necessarily call them fristen blankets because then it gets into that whole thing about it being his name but there's there is something possibly between or beyond a markovian blanket that would help denote what is going on with an active influence blanket where it's bi-directional um but this is this is great conversation we'll talk more about this obviously yep perfect um that's all that uh we need to i think note for the lobbying but yeah thanks for sharing about that these are the literal discussions in the terms discussions in edu every week these are like the fun things that we get to talk about so um to those in this jitsi many of whom were um just totally essential and non-fungible in their contributions to the lab as well as those who are watching the replay and who participated in every activity this year it was just an awesome year who could have thought that one year ago things would be where they are now the game is just beginning so we hope that everyone listening whatever their background or familiarity with active inference or other topics is they're just welcome to get involved any other last thoughts otherwise that's it for the fourth

quarterly roundtable meeting should i play the uh theme song again okay you i don't think you're gonna hear it on the uh you're not gonna hear it on the stream but so just it'll be quiet for 30 seconds oh wait i need to restart it without the little filter to do the voice okay so here we go theme song is going to start now and then i'll end the stream thanks and bye